

Online Learning Platform API

Overview

The Online Learning Platform API is a backend application developed using Spring Boot and MongoDB that enables the management of online educational resources. The system allows administrators and instructors to create and manage courses, students to enroll in courses, and instructors to assign learning materials and assignments.

The project demonstrates RESTful API development, MongoDB integration, CRUD operations, and entity relationship mapping using Spring Boot.

Objectives

- Manage students, instructors, courses, enrollments, and assignments.
- Perform complete CRUD operations on all entities.
- Implement relationships between collections.
- Provide REST APIs for data management.
- Test APIs using Postman and Swagger.
- Demonstrate scalable backend architecture using Spring Boot and MongoDB.

Features

Student Management

- Register students
- Update student profiles
- View enrolled courses
- Delete student records

Instructor Management

- Add instructors
- Update instructor information
- Assign instructors to courses
- Remove instructors

Course Management

- Create courses
- Update course details
- Delete courses
- View available courses

Enrollment Management

- Enroll students in courses
- View enrollment history

- Manage course participation

Assignment Management

- Create assignments
- Assign coursework to students
- Update assignment details
- Remove assignments

Technologies Used

Backend

- Java
- Spring Boot
- Spring Framework
- Spring Data MongoDB

Database

- MongoDB

API Documentation & Testing

- Swagger UI
- Postman

Build Tool

- Maven

Database Design

Collections

Students

- studentId
- name
- email
- contactNumber
- enrolledCourses

Instructors

- instructorId
- name
- specialization
- email
- assignedCourses

Courses

- courseId
- title
- description
- duration

- instructorId

Enrollments

- enrollmentId
- studentId
- courseId
- enrollmentDate
- status

Assignments

- assignmentId
- title
- description
- dueDate
- courseId

Relationships

One-to-Many

- Instructor → Courses
- Course → Assignments

Many-to-One

- Courses → Instructor

- Enrollments → Student
- Enrollments → Course

Many-to-Many

- Students ↔ Courses (through Enrollments)

CRUD Operations

The following operations are supported for all entities:

- Create
- Read
- Update
- Delete

Entities:

- Students
- Instructors
- Courses
- Enrollments
- Assignments

Project Structure

src/main/java

├── controller

└— service

└— repository

└— model

└— dto

└— config

└— exception

└— OnlineLearningPlatformApplication.java

REST API Endpoints

Student APIs

Method	Endpoint	Description
POST	/api/students	Create Student
GET	/api/students	Get All Students
GET	/api/students/{id}	Get Student By ID
PUT	/api/students/{id}	Update Student
DELETE	/api/students/{id}	Delete Student

Instructor APIs

Method	Endpoint	Description
POST	/api/instructors	Create Instructor

Method	Endpoint	Description
GET	/api/instructors	Get All Instructors
GET	/api/instructors/{id}	Get Instructor By ID
PUT	/api/instructors/{id}	Update Instructor
DELETE	/api/instructors/{id}	Delete Instructor

Course APIs

Method	Endpoint	Description
POST	/api/courses	Create Course
GET	/api/courses	Get All Courses
GET	/api/courses/{id}	Get Course By ID
PUT	/api/courses/{id}	Update Course
DELETE	/api/courses/{id}	Delete Course

Enrollment APIs

Method	Endpoint	Description
POST	/api/enrollments	Enroll Student
GET	/api/enrollments	Get All Enrollments
PUT	/api/enrollments/{id}	Update Enrollment

Method Endpoint

Description

DELETE /api/enrollments/{id} Delete Enrollment

Assignment APIs

Method Endpoint

Description

POST /api/assignments Create Assignment

GET /api/assignments Get All Assignments

GET /api/assignments/{id} Get Assignment By ID

PUT /api/assignments/{id} Update Assignment

DELETE /api/assignments/{id} Delete Assignment

Installation and Setup

Clone Repository

```
git clone https://github.com/your-username/online-learning-platform-api.git
```

```
cd online-learning-platform-api
```

Configure MongoDB

Update the application.properties file:

```
spring.data.mongodb.uri=mongodb://localhost:27017/  
online_learning_db
```

Build the Project

mvn clean install

Run the Application

mvn spring-boot:run

The application will start on:

<http://localhost:8080>

API Documentation

Swagger UI:

<http://localhost:8080/swagger-ui/index.html>

Testing

Use Postman or Swagger UI to test all CRUD operations and entity relationships.

Future Enhancements

- JWT Authentication and Authorization
- Role-Based Access Control (Admin, Instructor, Student)
- Course Completion Certificates
- Quiz and Examination Module
- Video Lecture Management
- Progress Tracking Dashboard
- Notification System

- File Upload and Resource Sharing
- Course Reviews and Ratings

Learning Outcomes

- Spring Boot REST API Development
- MongoDB Database Integration
- CRUD Operations
- Collection Relationship Mapping
- Layered Architecture Implementation
- API Documentation with Swagger
- Backend Project Design Principles

Conclusion

The Online Learning Platform API is a comprehensive backend project that demonstrates modern web application development using Spring Boot and MongoDB. It showcases entity relationships, RESTful services, database operations, and scalable backend architecture suitable for real-world e-learning platforms.